

SIMATS ENGINEERING
CO2_ASSESSMENT TOOL3_Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I P. Guru/192521010 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P. Guru

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.
2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.
3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge

servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

1. **A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.**

ANS:

- Many businesses experience sudden increases in website traffic during special events such as festive sales, flash sales, Black Friday, or New Year offers. If the server cannot handle the increased number of users, the website becomes slow or even crashes, resulting in loss of customers and revenue. Cloud computing solves this problem by providing Auto-Scaling and Load Balancing, which automatically adjust computing resources according to demand while ensuring that incoming traffic is distributed efficiently.

Cloud Auto-Scaling

Auto-scaling is a cloud feature that automatically increases or decreases computing resources such as virtual machines or application servers based on current workload. Instead of purchasing expensive hardware for peak demand, organizations can dynamically allocate resources only when required.

Working of Auto-Scaling

1. During normal business hours, only a few cloud servers are active.
2. When user traffic suddenly increases, monitoring services detect high CPU utilization, memory usage, or network traffic.
3. Auto-scaling automatically launches additional virtual machines or containers.
4. During low traffic periods, unnecessary servers are removed automatically.
5. This ensures efficient resource utilization while minimizing operational costs.

Advantages of Auto-Scaling

- Handles sudden traffic spikes automatically.
- Prevents application crashes.
- Optimizes cloud resource utilization.
- Reduces infrastructure costs through pay-as-you-use pricing.
- Improves business continuity during peak demand.

Cloud Load Balancing

Load balancing distributes incoming user requests across multiple servers so that no single server becomes overloaded.

Working Principle

Instead of sending all users to one server, the load balancer acts as an intelligent traffic controller.

- User requests first reach the load balancer.
- The load balancer checks server availability and workload.
- Requests are forwarded to the least busy or healthiest server.
- If one server fails, traffic is automatically redirected to healthy servers.

Common load-balancing algorithms include:

- Round Robin
- Least Connections
- Weighted Round Robin
- IP Hash

Benefits of Load Balancing

- Prevents server overload.
- Ensures high availability.
- Improves response time.
- Eliminates single points of failure.
- Supports fault tolerance.

Reducing Downtime and Improving User Experience

Downtime during sales can result in customer dissatisfaction and financial loss. Combining auto-scaling with load balancing ensures continuous service.

Reduction of Downtime

- Additional servers are launched before existing servers become overloaded.
- Failed servers are automatically removed from service.
- User requests are redirected to healthy servers.
- Applications remain accessible even during hardware failures.

Improved User Experience

Customers experience:

- Faster page loading
- Quick product search

- Smooth checkout process
- Reduced waiting time
- High website availability

A smooth user experience increases customer satisfaction, trust, and sales.

Role of Monitoring Tools

Monitoring tools continuously observe cloud infrastructure and application performance.

They collect metrics such as:

- CPU utilization
- Memory usage
- Disk usage
- Network traffic
- Request latency
- Response time
- Error rates

Examples of monitoring tools include:

- Amazon CloudWatch
- Microsoft Azure Monitor
- Google Cloud Monitoring
- Prometheus
- Grafana

Functions of Monitoring Tools

- Detect abnormal traffic spikes.
- Trigger auto-scaling policies.
- Generate alerts for administrators.
- Identify performance bottlenecks.
- Produce reports for system optimization.

Without monitoring, auto-scaling decisions cannot be made efficiently.

Real-World E-Commerce Scenario

Consider **Amazon** during the **Great Indian Festival Sale**.

During the sale:

- Millions of users visit the website simultaneously.
- Customers browse products, compare prices, add items to carts, and complete purchases.

How Cloud Services Help

1. Monitoring detects increased CPU and network traffic.
2. Auto-scaling launches additional cloud servers.
3. Load balancers distribute customer requests evenly across all servers.
4. If one server becomes unavailable, traffic automatically shifts to another healthy server.
5. After the sale ends, extra servers are removed automatically to reduce costs.

As a result:

- Customers experience fast page loading.
- Orders are processed without interruption.
- Payment gateways remain responsive.
- The company avoids revenue loss due to server crashes.

Advantages:

Feature	Benefit
Auto-Scaling	Automatically adjusts resources according to demand
Load Balancing	Distributes traffic evenly across servers
Monitoring	Detects workload changes and triggers scaling
High Availability	Minimizes downtime
Cost Efficiency	Pay only for resources used
Fault Tolerance	Redirects traffic when servers fail
Performance	Faster response time and better user experience

2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.

ANS:

- Modern organizations generate enormous amounts of digital data every day, including customer records, financial transactions, employee information, business reports, and application databases. Protecting this data from hardware failures, cyberattacks, natural disasters, or accidental deletion is a major challenge. Cloud storage provides a reliable, scalable, and secure solution for storing large volumes of backup data. Cloud providers such as Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform (GCP) offer highly durable storage systems with built-in redundancy, encryption, and disaster recovery capabilities.

Cloud Storage Solutions

Cloud storage is a service that allows organizations to store data on remote servers managed by cloud providers instead of maintaining physical storage devices in their own data centers. Data is stored across multiple geographically distributed servers and can be accessed through the internet whenever required.

Popular cloud storage services include:

- Amazon S3 (Simple Storage Service)
- Microsoft Azure Blob Storage
- Google Cloud Storage

These services are designed to handle petabytes of data while maintaining high availability and security.

Durability and Data Redundancy

One of the biggest advantages of cloud storage is its ability to protect data against failures using durability and redundancy mechanisms.

1. Data Durability

Durability refers to the ability of the cloud storage system to preserve data without loss over a long period.

Cloud providers achieve high durability by:

- Storing multiple copies of data.
- Continuously checking data integrity.

- Automatically repairing corrupted or damaged files.
- Replacing failed storage devices without affecting users.

Many cloud providers guarantee durability of **99.999999999% (11 nines)**, meaning the probability of losing stored data is extremely low.

Benefits

- Protects against hardware failures.
- Prevents permanent data loss.
- Ensures long-term preservation of backup files.

2. Data Redundancy

Redundancy means storing multiple copies of the same data in different storage devices or different geographical locations.

For example:

- File A is stored in Data Center 1.
- Another copy is stored in Data Center 2.
- A third copy is stored in Data Center 3.

If one data center becomes unavailable due to power failure, flood, earthquake, or hardware malfunction, data can still be retrieved from another location.

Advantages of Redundancy

- High availability
- Fault tolerance
- Disaster protection
- Continuous data access
- Improved reliability

Encryption for Data Security

Organizations often store confidential information such as customer details, payroll records, financial statements, and intellectual property. Encryption protects this sensitive data from unauthorized access.

Encryption converts readable information (plaintext) into an unreadable format (ciphertext) using cryptographic algorithms.

Only users possessing the correct encryption key can decrypt and access the original information.

Types of Encryption

1. Encryption at Rest

- Protects data stored inside cloud storage.
- Even if storage disks are stolen, the data remains unreadable.
- Common encryption standards include **AES-256**.

2. Encryption in Transit

- Protects data while it travels between the user and cloud servers.
- Uses secure communication protocols such as **SSL/TLS**.
- Prevents attackers from intercepting sensitive information.

Benefits of Encryption

- Prevents unauthorized access.
- Protects confidential business data.
- Ensures customer privacy.
- Supports regulatory compliance.

Access Control for Secure Backup Data

Encryption alone is not sufficient. Organizations must also control who can access the stored backups.

Access control determines which users are allowed to view, modify, or delete cloud data.

Common access control methods include:

Identity and Access Management (IAM)

Allows administrators to assign permissions based on user roles.

Example:

- Administrator – Full access
- Manager – Read and write access
- Employee – Read-only access
- Guest – No access

Multi-Factor Authentication (MFA):

Users must verify their identity using:

- Password
- Mobile OTP
- Authentication app
- Fingerprint or Face ID

Role-Based Access Control (RBAC):

Permissions are assigned according to job responsibilities.

Example:

- HR department can access employee records.
- Finance department can access accounting backups.
- IT administrators manage storage infrastructure.

Benefits:

- Prevents unauthorized access.
- Reduces insider threats.
- Improves accountability.
- Enhances data security.

Advantages of Cloud Storage:

Feature	Benefit
High Durability	Protects data from permanent loss
Data Redundancy	Stores multiple copies across locations
Encryption	Secures sensitive information
Access Control	Restricts unauthorized users
Scalability	Easily increase or decrease storage
Cost Efficiency	Pay only for storage used

3. A mobile app company plans to deploy its application globally. Explain how Content Delivery Networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

ANS:

- As mobile applications become popular worldwide, users expect fast loading times and uninterrupted services regardless of their geographical location. If all user requests are handled by a single central server, users located far away experience high latency, slow response times, buffering, and poor application performance. Cloud computing addresses this challenge through Content Delivery Networks (CDNs) and Edge Servers. CDNs distribute content across multiple servers located in different regions of the world, allowing users to access data from the nearest server instead of the origin server. This significantly improves application speed, reliability, and user satisfaction. Major cloud providers such as Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform (GCP) offer global CDN services that ensure high availability and seamless content delivery.

Content Delivery Network (CDN):

A **Content Delivery Network (CDN)** is a distributed network of servers strategically located across different countries and regions. These servers store copies (cached versions) of frequently accessed content such as:

- Images
- Videos
- Audio files
- JavaScript files
- CSS files
- Mobile app updates
- Website content

Instead of every user requesting data from the main server, the CDN delivers content from the nearest edge server, reducing the distance data must travel.

How CDNs Improve Performance?

When a user opens a mobile application, the request first reaches the nearest CDN server.

Working Process:

1. The mobile user requests content.
2. The CDN identifies the nearest edge server.

3. If the requested content is available in the cache, it is delivered immediately.
4. If the content is unavailable, the edge server retrieves it from the origin server.
5. The content is cached for future requests.

This process minimizes network congestion and significantly improves response speed.

Benefits of CDN:

1. Reduced Latency

Latency is the time taken for data to travel between the user and the server.

CDNs reduce latency because:

- Data travels a shorter distance.
- Requests are handled by nearby servers.
- Network delays are minimized.

Result:

- Faster application response.
- Quick content loading.
- Smooth user interaction.

2. Faster Content Delivery

Frequently accessed files are stored in CDN cache.

Benefits include:

- Faster image loading.
- Instant video playback.
- Reduced buffering.
- Improved website speed.

3. Lower Server Load

Instead of handling millions of user requests, the origin server only serves requests when cached data is unavailable.

This reduces:

- CPU utilization
- Network traffic

- Server workload

As a result, the application remains stable during heavy traffic.

4. Higher Availability

If one CDN server experiences failure, another nearby edge server automatically delivers the content.

This provides:

- High availability
- Fault tolerance
- Continuous service

Edge Servers and User Experience

Edge servers are geographically distributed servers positioned close to users.

Their primary function is to store cached copies of frequently accessed content.

Advantages of Edge Servers

1. Faster Response Time

Users receive data from nearby servers instead of distant data centers.

For example:

- A user in India receives content from an Indian edge server instead of a server located in the United States.

This significantly improves application speed.

2. Reduced Network Congestion

Since requests are distributed among multiple edge servers:

- Internet backbone traffic decreases.
- Packet loss is minimized.
- Data transfer becomes more efficient.

3. Better Streaming Quality

Edge servers support:

- HD video streaming
- 4K video playback

- Live broadcasts
- Music streaming

Users experience:

- Less buffering
- Stable playback
- Higher video quality

4. Improved Gaming Experience

Online games require very low latency.

Edge servers help by:

- Reducing response delay.
- Synchronizing game data quickly.
- Providing smoother multiplayer gameplay.
- Minimizing lag.

Advantages of CDN and Edge Servers:

Feature	Benefit
CDN	Delivers content from nearby servers
Edge Servers	Reduce latency and improve speed
Caching	Stores frequently accessed content
Global Availability	Provides uninterrupted worldwide access
Load Balancing	Distributes user traffic efficiently
Auto Scaling	Handles sudden increases in users
Fault Tolerance	Maintains service during server failures
Improved User Experience	Faster loading and reduced buffering

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.

ANS:

- Cloud computing provides advanced security features and compliance frameworks that help financial organizations securely store, process, and manage their data. Cloud providers such as Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform (GCP) offer built-in services for regulatory compliance, auditing, encryption, logging, monitoring, and data sovereignty, enabling organizations to meet strict legal and industry requirements.

Regulatory Compliance in Cloud Computing:

Regulatory compliance means following government laws, industry regulations, and security standards while storing and processing sensitive information.

Cloud providers regularly undergo independent security audits and obtain internationally recognized certifications to demonstrate compliance with global standards.

Some common compliance standards include:

- **ISO 27001** – Information Security Management
- **SOC 1, SOC 2, and SOC 3** – Security and operational controls
- **PCI-DSS** – Security standard for credit/debit card payments
- **GDPR** – Protection of personal data (European Union)
- **HIPAA** – Healthcare information protection
- **RBI Guidelines** – Cloud usage recommendations for Indian financial institutions

These certifications assure organizations that cloud infrastructure follows strict security and operational practices.

Benefits of Regulatory Compliance

- Protects confidential customer information.
- Reduces legal and financial risks.
- Ensures adherence to government regulations.
- Builds customer trust and confidence.
- Simplifies external audits.

Cloud Auditing:

Auditing is the process of continuously recording and reviewing all activities performed within the cloud environment.

Cloud providers automatically maintain detailed audit logs that record:

- User login attempts
- File access history
- Configuration changes
- Data uploads and downloads
- Transaction activities
- Administrative actions
- Security policy modifications

These records help organizations investigate suspicious activities and demonstrate compliance during inspections.

Advantages of Auditing

- Detects unauthorized activities.
- Provides complete accountability.
- Supports forensic investigations.
- Helps identify policy violations.
- Simplifies compliance reporting.

Role of Logging in Security:

Logging refers to automatically recording every important event that occurs within the cloud system.

Typical logged events include:

- Successful and failed login attempts
- Password changes
- Database access
- File modifications

- API requests
- Financial transactions
- User activities

Importance of Logging:

- Tracks user behaviour.
- Detects unauthorized access attempts.
- Provides evidence during investigations.
- Helps identify cyberattacks.
- Supports regulatory compliance.

Without proper logging, it becomes difficult to determine how a security incident occurred.

Role of Monitoring:

Monitoring continuously observes the health, performance, and security of cloud resources in real time.

Monitoring systems track:

- CPU utilization
- Memory usage
- Network traffic
- Failed login attempts
- Suspicious IP addresses
- Malware detection
- Unusual transaction patterns
- System availability

Cloud monitoring tools automatically generate alerts whenever abnormal activities are detected.

Examples include:

- Amazon CloudWatch
- Azure Monitor
- Google Cloud Monitoring

Benefits of Monitoring

- Detects security threats quickly.
- Prevents system failures.
- Supports automatic incident response.
- Improves overall system reliability.
- Ensures continuous service availability.

Role of Encryption:

Encryption is one of the most important security mechanisms used in cloud computing.

It converts readable information (**plaintext**) into unreadable data (**ciphertext**) using cryptographic algorithms.

Only authorized users with the correct encryption key can decrypt the information.

1. Encryption at Rest

Data stored in cloud databases or storage systems is encrypted using algorithms such as **AES-256**.

Benefits:

- Protects stored customer information.
- Prevents unauthorized access if storage devices are compromised.
- Secures backup files.

2. Encryption in Transit

When data moves between users and cloud servers, secure communication protocols such as **SSL/TLS** encrypt the transmission.

Benefits:

- Prevents data interception.
- Protects online banking sessions.
- Secures payment transactions.
- Prevents man-in-the-middle attacks.

Data Sovereignty:

Data sovereignty means that an organization's data must remain stored and processed according to the laws of the country where it originates.

Many governments require sensitive financial information to remain within national borders.

Cloud providers support data sovereignty by allowing organizations to choose the geographic region where data will be stored.

For example:

- An Indian bank can choose the **Mumbai** or **Hyderabad** cloud region.
- Customer financial records remain stored within India.
- Data processing follows Indian laws and regulatory guidelines.

How Companies Ensure Data Sovereignty

- Select appropriate cloud regions.
- Restrict cross-border data transfers.
- Apply region-specific backup policies.
- Use encryption and access controls.
- Conduct regular compliance audits.

Benefits

- Compliance with national regulations.
- Improved customer trust.
- Reduced legal complications.
- Better control over sensitive information.

5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

ANS:

- Traditional deployment methods using Virtual Machines (VMs) often consume significant system resources because each VM runs its own operating system. To overcome these limitations, organizations increasingly use containerization, where applications and their dependencies are packaged into lightweight containers. Containers are faster, more portable, and require fewer resources than virtual machines. To manage large numbers of containers, orchestration platforms such as Kubernetes automate deployment, scaling, networking, and monitoring. Cloud providers such as Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform (GCP) offer managed Kubernetes services, making containerized applications easier to deploy globally.

Containerization

A **container** is a lightweight software package that includes:

- Application code
- Runtime environment
- Libraries
- Configuration files
- Dependencies

Unlike virtual machines, containers share the host operating system kernel while remaining isolated from each other. This enables applications to run consistently across different environments.

Popular container platforms include:

- Docker
- Podman
- Containerd

Containers vs Virtual Machines

Although both containers and virtual machines provide isolated environments for applications, they differ significantly in architecture and performance.

Virtual Machines (VMs)

A virtual machine emulates a complete computer system.

Each VM contains:

- Guest Operating System
- Application
- Libraries
- Virtual Hardware

Because every VM runs its own operating system, it consumes more memory, storage, and CPU resources.

Advantages of Virtual Machines

- Strong isolation.
- Suitable for running different operating systems.
- Better compatibility for legacy applications.

Disadvantages

- High resource consumption.
- Slow startup time.
- Larger storage requirements.
- Higher maintenance cost.

Containers:

Containers share the host operating system kernel.

Each container includes only:

- Application
- Libraries
- Dependencies

Since a separate operating system is unnecessary, containers are lightweight and highly efficient.

Advantages of Containers:

- Very fast startup.
- Low memory usage.
- High performance.
- Efficient CPU utilization.
- Easy deployment.
- Supports rapid scaling.

Disadvantages

- Weaker isolation than virtual machines.
- Requires compatible host operating system.
- Security configuration must be carefully managed.

Performance and Resource Usage:

Containers provide significantly better performance because they eliminate the overhead of running multiple operating systems.

Performance Comparison:

Containers:

- Start within seconds.
- Consume less RAM.
- Require less storage.
- Faster application deployment.
- Higher server density.

Virtual Machines:

- Require several minutes to start.
- Consume large amounts of RAM.
- Require separate operating systems.
- Lower application density.
- Higher infrastructure costs.

As a result, organizations can run many more containers than virtual machines on the same physical server.

Kubernetes Orchestration:

When applications consist of hundreds or thousands of containers, manual management becomes difficult.

Kubernetes is an open-source container orchestration platform that automates container deployment and management.

Its primary responsibilities include:

- Container deployment
- Scheduling
- Load balancing
- Automatic scaling
- Health monitoring
- Self-healing
- Service discovery
- Rolling updates

How Kubernetes Simplifies Deployment

1. Automated Deployment

Developers define the desired number of containers.

Kubernetes automatically creates and deploys them on available servers.

2. Auto Scaling

When application traffic increases:

- Kubernetes automatically launches additional containers.

When traffic decreases:

- Unused containers are removed.

This ensures efficient resource utilization.

3. Self-Healing

If a container crashes:

- Kubernetes detects the failure.
- The failed container is automatically restarted or replaced.
- Users experience minimal downtime.

4. Load Balancing

Incoming user requests are distributed evenly across multiple containers.

Benefits include:

- Better performance.
- High availability.
- Prevention of container overload.

5. Rolling Updates

Developers can deploy new application versions without shutting down the entire application.

Benefits:

- Zero downtime.
- Easy rollback.
- Continuous service availability.

Portability Across Cloud Environments

One of the greatest advantages of containers is **portability**.

Since containers package the application along with all its dependencies, they behave consistently regardless of where they are executed.

A Docker container created on a developer's laptop can run without modification on:

- Amazon Web Services (AWS)
- Microsoft Azure

- Google Cloud Platform (GCP)
- Private cloud
- Hybrid cloud
- On-premises servers

This eliminates the common problem of "It works on my machine but not in production."

Benefits of Portability:

- Easy migration between cloud providers.
- Reduced vendor lock-in.
- Consistent application behavior.
- Faster software deployment.
- Simplified testing and development.

Microservices-Based Application Example:

Consider an **Online Food Delivery Application** such as **Swiggy** or **Zomato**.

Instead of building the entire application as one large system, it is divided into multiple independent **microservices**.

Example Microservices

- User Service – Manages user registration and login.
- Restaurant Service – Displays restaurant details and menus.
- Order Service – Handles food orders.
- Payment Service – Processes online payments.
- Delivery Service – Tracks delivery partners.
- Notification Service – Sends SMS, email, and app notifications.

Each service runs inside its own Docker container.

Kubernetes manages all these containers by:

- Deploying services automatically.
- Scaling only the busy services.
- Restarting failed containers.

- Balancing traffic across multiple instances.

For example, during lunch hours, the **Order Service** may receive heavy traffic. Kubernetes automatically increases the number of Order Service containers without affecting the other services.

This ensures smooth operation, high availability, and efficient use of resources.

Advantages of Containerization:

Feature	Benefit
Lightweight Architecture	Low memory and CPU usage
Fast Startup	Containers start in seconds
High Performance	Better application efficiency
Kubernetes Orchestration	Automates deployment and scaling
Self-Healing	Automatically replaces failed containers
Portability	Runs consistently across different cloud platforms
Scalability	Easily handles increasing workloads
Microservices Support	Independent deployment and management of services

